

Lecture 2 Introduction to Linux Notes

1. What is an Operating System?

An operating system is the main software that helps a computer run. It manages the computer's hardware and allows the user and other programs to interact with the computer.

Some examples are:

- Windows
- macOS
- Linux
- Android
- iOS

2. What is a Kernel?

The kernel is basically the main part of an operating system. It connects the software to the computer's hardware and manages things like the CPU, memory, devices, and running programs.

3. Which Other Parts Aside From the Kernel Identify an OS?

The kernel is important, but it is not the only part of an operating system. Other parts include:

- System utilities
- Device drivers
- System libraries
- User interface
- Applications
- Package management tools

All of these parts work together to make the operating system usable.

4. What is Linux?

Linux is an open-source operating system kernel. It was created by Linus Torvalds in 1991.

Linux is used as the foundation for many different operating systems. These are called Linux distributions.

5. What is a Linux Distribution?

A Linux distribution is a complete operating system that uses the Linux kernel.

A distribution usually includes the kernel, system tools, applications, libraries, and a way to install and manage software.

Some examples are:

- Debian

- Ubuntu
- Fedora
- Arch Linux
- Linux Mint

6. List at Least 4 Linux Characteristics

Some characteristics of Linux are:

- **Open source** - The source code is available and can be changed and shared under its license.
- **Multiuser** - More than one user can have access to the system.
- **Multitasking** - Linux can run multiple programs and processes at the same time.
- **Portable** - Linux can work on many different types of computers and hardware.
- **Secure** - Linux uses permissions and other security features to protect files and resources.
- **Customizable** - Users can change many parts of the system and choose different software.

7. What is Debian?

Debian is a free and open-source Linux distribution. It is developed by a community of people who work on the project.

Debian is known for being stable and having a large amount of available software. Ubuntu is also based on Debian.

8. List and Define the Different Types of Licensing Agreements

Proprietary Software

Proprietary software is owned by a person or company. The user usually has limits on copying, changing, or sharing the software.

Open Source Software

Open source software makes its source code available. Depending on the license, users can study it, modify it, and share it.

Free Software

Free software focuses on the freedoms users have with the software. Users should be able to use, study, modify, and share it.

Freeware

Freeware is software that can usually be used without paying for it. However, the source code is not necessarily available to users.

Shareware

Shareware is software that lets people try it before buying it. Some features or continued use may require payment.

9. What is Free Software? Define the 4 Freedoms.

Free Software is software that gives users the freedom to use, study, change, and share the software.

The four freedoms are:

1. **Freedom 0** - The freedom to run the program for any purpose.
2. **Freedom 1** - The freedom to study how the program works and change it. You need access to the source code for this.
3. **Freedom 2** - The freedom to share copies of the program with other people.
4. **Freedom 3** - The freedom to share modified versions so other people can benefit from the changes.

10. What is Virtualization?

Virtualization is a way of using one physical computer to run one or more virtual computers.

A virtual machine can have its own operating system, storage, memory, and applications. A program called a hypervisor manages the virtual machines and lets them use the physical computer's resources.

For example, I am using VirtualBox to run Debian Linux as a virtual machine on my Mac.